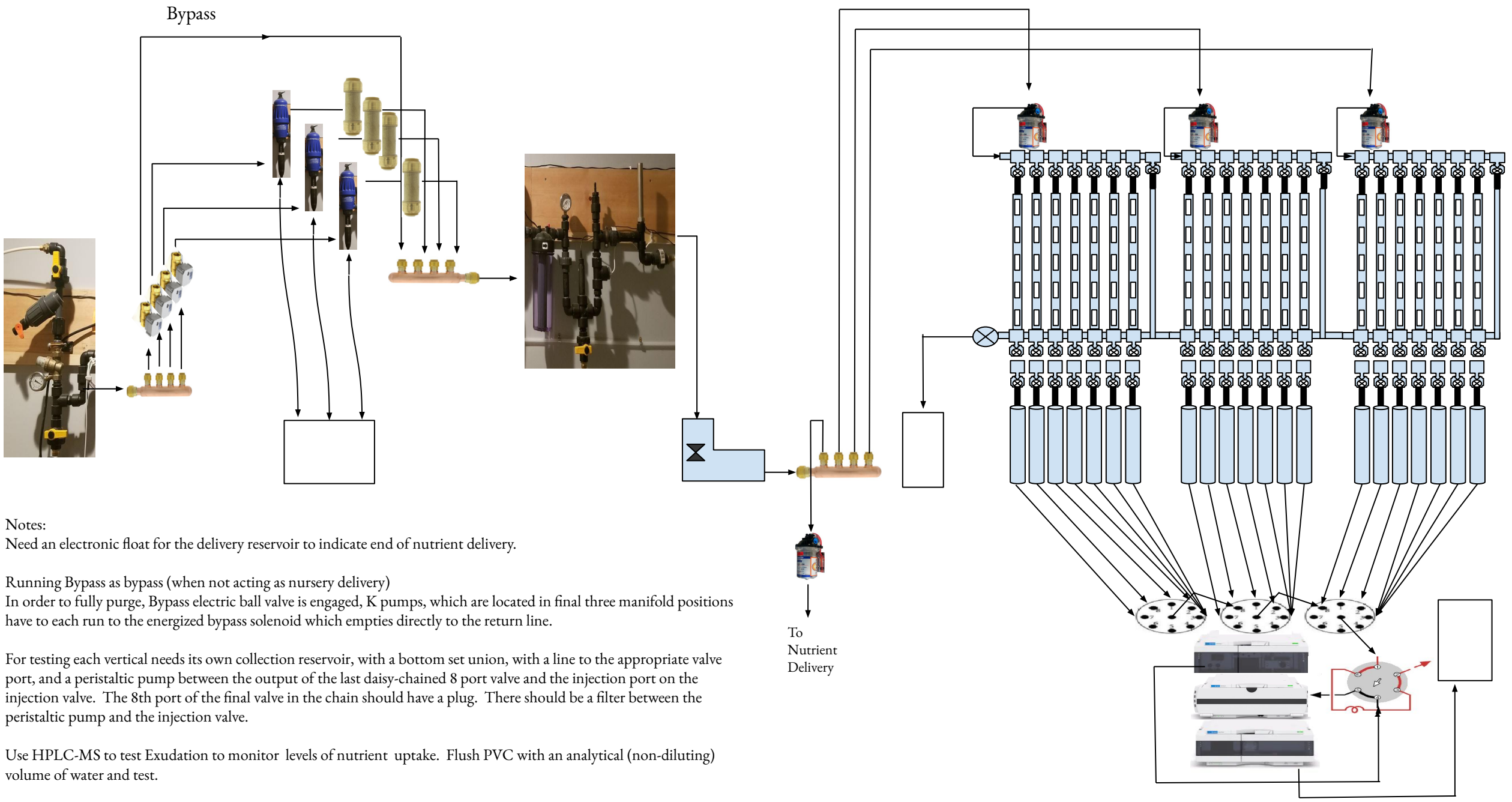

Notes:
Need an electronic float for the delivery reservoir to indicate end of nutrient delivery.

Running Bypass as bypass (when not acting as nursery delivery)
In order to fully purge, Bypass electric ball valve is engaged, K pumps, which are located in final three manifold positions have to each run to the energized bypass solenoid which empties directly to the return line.

For testing each vertical needs its own collection reservoir, with a bottom set union, with a line to the appropriate valve port, and a peristaltic pump between the output of the last daisy-chained 8 port valve and the injection port on the injection valve. The 8th port of the final valve in the chain should have a plug. There should be a filter between the peristaltic pump and the injection valve.

Use HPLC-MS to test Exudation to monitor levels of nutrient uptake. Flush PVC with an analytical (non-diluting) volume of water and test.